

# Self-Organized Criticality: Sandpile Model on Undirected Graph, Influence of Topology and Dissipation Rules on Avalanches

## 1. Introduction

Natural systems with high degree of freedom such as earthquakes often exhibit a tendency to reach a critical state. This phenomenon is called self-organized criticality. These systems most famously need no fine-tuning in order to reach the critical state, rather it is an inherent property of theirs. Systems in a critical state are very sensitive to inputs and this leads to its unpredictability.

The concept of SOC was first discovered in 1987 by Bak, Tang and Wiesenfeld and its property demonstrated on a sandpile model [1]. This model was later generalized from square lattice to arbitrary graph. That generalization is important because it allows us to study the models behaviour for different graph topologies.

## 2. Background

### 2.1. Sandpile Model

The original sandpile model presents a simple chessboard and a falling sand grains. Each grain fall on some chess square determined by a random distribution. If the number of grains on any square reach four, the sandpile topples and the four grains fall on adjoint squares, possibly causing additional squares to topple. In case the square does not have four adjacent squares (ie it is on the edge of chessboard) the remaining grains fall off the board and are therefore removed from the model.

This model can be expressed as an undirected graph, a square lattice of the size  $8 \times 8$ . This however is not a correct representation until an additional vertex is introduced. The often called sink vertex have an important property that it cannot be toppled. When the sink is connected to every vertex on the edge of the square lattice (twice to corner vertices) the graph representation becomes valid. This is foundational observation, for it allows us to use any underlying graph for the sandpile model.

TODO: sink is needed for the model to not end-up in infinite loop

### 2.2. Self-Organized Criticality

TODO: critical state, power-law

### 2.3. Graph Theory

In order to explore the behaviour of the sandpile model on a graph, we must establish these concepts. First is *vertex degree*  $d(v_i)$  which is defined as equal to the number of vertices connected to  $v_i$  via an edge.

*Path length* describes the distance between two vertexes for a given path. The path length is equal to the number of vertices in between the first and last vertex on a path. The *shortest path length* refers to a distance between two vertexes such as any other path between them produces at least the same path length. It's useful to also define an *average shortest path length*, an average of shortest paths between any two vertexes in a graph.

TODO: small-world network properties (the previous needed for that)

### 3. Model Definition

#### 3.1. Sandpile on Graphs

As discussed in *Sandpile Model* (page 1) the existence of a sink vertex is necessary for any graph representation of a sandpile model.

### Bibliography

- [1] P. Bak, C. Tang, and K. Wiesenfeld, “Self-organized criticality: An explanation of the  $1/f$  noise,” *Physical Review Letters*, vol. 59, no. 4, pp. 381–384, July 1987, doi: 10.1103/PhysRevLett.59.381.